

Lecture 8

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- The log determinant function - definitions, importance, and derivatives
- Semidefinite programming example
- Hessians - definitions for vector-variate and matrix-variate functions
- Chain rules for Hessians

The log determinant function

An important example of a multivariate function that's not so easy to differentiate is the log determinant function

$$f : \mathbb{S}_{++}^n \rightarrow \mathbb{R} \quad \text{given by} \quad X \mapsto \log \det X,$$

This is well-defined since eigenvalues of X and thus $\det X$, which is the product of the eigenvalues, is always positive.

This function is very important in the field of semidefinite programming (SDP), which is probably the most important optimization problem in modern convex optimization.

It is also very important in machine learning and statistics because when you take log of the Gaussian density

$$\frac{1}{(2\pi)^{n/2} \det(\Sigma)^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \mu)^\top \Sigma (\mathbf{x} - \mu) \right],$$

which you have to when you write down a log likelihood function, you get

$$(\mathbf{x} - \mu)^\top \Sigma (\mathbf{x} - \mu) + \log \det(\Sigma)$$

after dropping the (inconsequential) constants. Doing maximum likelihood then requires you to minimize a function involving log det.

Finding the gradient of the log determinant function

We claim that

$$\nabla \log \det(X) = X^{-1}$$

This is consistent with the $n = 1$ case: Here $\det([x]) = x$, so $\log \det x = \log x$, and the derivative is $1/x$. To show (8.1), we need to recall a few basic things:

For an invertible matrix R , taking transpose and taking inverse are interchangeable, i.e.,

$$(R^{-1})^{\top} = (R^{\top})^{-1}$$

and instead of preferring one over another, we just write $R^{-\top}$;

The trace and determinant of $A \in \mathbb{R}^{n \times n}$ satisfy

$$\operatorname{tr}(A) = \lambda_1 + \cdots + \lambda_n, \quad \det(A) = \lambda_1 \cdots \lambda_n$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A .

Proof of the gradient of the log determinant function

Fix $X \in \mathbb{S}_{++}^n$, and let $H \in \mathbb{S}^n$.

If it is true that $\nabla \log \det(X) = X^{-1}$, then the derivative of the log determinant function at X in the direction H is given by $\text{tr}(X^{-1}H)$, and the following limit expression must hold:

$$\lim_{H \rightarrow 0} \frac{|\log \det(X + H) - \log \det(X) - \text{tr}(X^{-1}H)|}{\|H\|} = 0$$

To show this, we'll need to know some things about $X^{-1}H$.

The eigenvalues of $X^{-1}H$ are real

We know that the eigenvalues of X are real because X is symmetric, and we know that they are positive because X is positive definite.

We will need to know in our proof that the eigenvalues of $X^{-1}H$ are real. Unfortunately, $X^{-1}H$ is not necessarily symmetric (even though X^{-1} and H are), so we can't use the fact that the eigenvalues of a symmetric matrix are real.

Instead, we are going to find a specific matrix that is *similar* to $X^{-1}H$ and show that it is symmetric. Then, since similar matrices have the same eigenvalues, we will know that the eigenvalues of $X^{-1}H$ are real.

(Remember, two matrices A and B are similar if there exists an invertible matrix R such that $B = R^{-1}AR$.)

From the last lecture, we know that there exists a matrix $R \in \text{GL}(n)$ (that is, R is invertible) such that $X = RR^\top$.

We notice that $R^{-1}HR^{-\top}$ is symmetric. (Why?)

So all we need to do is show that $X^{-1}H$ and $R^{-1}HR^{-\top}$ are similar matrices.

$$R^\top (X^{-1}H) R^{-\top} = R^\top (R^{-\top} R^{-1}H) R^{-\top} = R^{-1}HR^{-\top}$$

Since they are similar, they have the same eigenvalues. Since $R^{-1}HR^{-\top}$ is symmetric, those eigenvalues must be real. Call them $\lambda_1, \dots, \lambda_n$.

Remember, our limit expression is

$$\lim_{H \rightarrow 0} \frac{|\log \det(X + H) - \log \det(X) - \text{tr}(X^{-1}H)|}{\|H\|} = 0$$

Let's focus on the first two terms in the numerator. Just using properties of logarithms, we can rewrite these terms as

$$\begin{aligned}\log \det(X + H) - \log \det(X) &= \log[\det(X + H)/\det(X)] \\ &= \log \det[X^{-1}(X + H)] \\ &= \log \det(I + X^{-1}H)\end{aligned}$$

To find the determinant of $I + X^{-1}H$, we will use the fact that the eigenvalues of any matrix $I + A$ are just one plus the eigenvalues of A .

Since the determinant of a matrix is the product of its eigenvalues the determinant of $I + X^{-1}H$ is just the product of which are $(1 + \lambda_1), \dots, (1 + \lambda_n)$.

So we have

$$\begin{aligned}\log \det(X + H) - \log \det(X) &= \\ \log \det(I + X^{-1}H) &= \log [(1 + \lambda_1) \times \dots \times (1 + \lambda_n)]\end{aligned}$$

And then knowing that the log of a product is the sum of the logs, we get

$$\log \det(X + H) - \log \det(X) = \log(1 + \lambda_1) + \dots + \log(1 + \lambda_n).$$

Checking that the terms $1 + \lambda_i$ are positive

If any of the terms $1 + \lambda_i$ are negative, then we can't take log of them.

But remember what the λ_i are: they are the eigenvalues of $X^{-1}H$. As $H \rightarrow 0$, it makes intuitive sense that these eigenvalues will become small. But let's show that formally:

Suppose we have the eigenvector $\mathbf{v}_i$ of $X^{-1}H$ with eigenvalue λ_i , chosen to be a unit vector. Then we have

$$\begin{aligned}X^{-1}H\mathbf{v}_i &= \lambda_i\mathbf{v}_i \\ \|X^{-1}H\mathbf{v}_i\| &= |\lambda_i| \|\mathbf{v}_i\| \\ \|X^{-1}H\mathbf{v}_i\| &= |\lambda_i|\end{aligned}$$

Now we use the submultiplicativity of the norm, and the fact that $\|\mathbf{v}_i\| = 1$, to get

$$|\lambda_i| = \|X^{-1}H\mathbf{v}_i\| \leq \|X^{-1}H\| \|\mathbf{v}_i\| = \|X^{-1}H\|.$$

$$|\lambda_i| = \|X^{-1}H\mathbf{v}_i\| \leq \|X^{-1}H\| \|\mathbf{v}_i\| = \|X^{-1}H\|.$$

Using submultiplicativity a second time, we get

$$|\lambda_i| \leq \|X^{-1}H\| \leq \|X^{-1}\| \|H\|.$$

So as $H \rightarrow 0$, all eigenvalues of $X^{-1}H$ must go to zero too. Thus even if a λ_i is negative, it will eventually be small enough so that $1 + \lambda_i > 0$.

$$\begin{aligned}
 & \frac{|\log \det(X+H) - \log \det(X) - \text{tr}(X^{-1}H)|}{\|H\|} \\
 & \leq \frac{|\log \det(X+H) - \log \det(X) - \text{tr}(X^{-1}H)| \|X^{-1}\|}{\|X^{-1}H\|} && \text{by (8.4)} \\
 & = \frac{|\log(1+\lambda_1) + \dots + \log(1+\lambda_n) - \lambda_1 - \dots - \lambda_n|}{\|X^{-1}H\|} \times \|X^{-1}\| && \text{by (8.3),} \\
 & = \|X^{-1}\| \sum_{i=1}^n \frac{\frac{|\log(1+\lambda_i) - \lambda_i|}{\|X^{-1}H\|}}{\|X^{-1}H\|} \\
 & \leq \|X^{-1}\| \sum_{i=1}^n \frac{|\log(1+\lambda_i) - \lambda_i|}{|\lambda_i|} && \text{by (8.5)}
 \end{aligned}$$

The last expression goes to zero because, as we saw above, as $H \rightarrow 0$, all $\lambda_i \rightarrow 0$, and

$$\lim_{\lambda \rightarrow 0} \frac{\log(1+\lambda) - \lambda}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{1}{1+\lambda} - 1 = 0$$

by l'Hôpital.

Example: semidefinite programming

Let $A_0, A_1, \dots, A_n \in \mathbb{S}^p$ and

$$\Omega = \{\mathbf{x} \in \mathbb{R}^n : A_0 + x_1 A_1 + \dots + x_n A_n \succ 0\}.$$

(Note that an inequality of the form

$$A_0 + x_1 A_1 + \dots + x_n A_n \succ 0 \quad \text{or} \quad \succeq 0$$

is called a linear matrix inequality.)

Let $f : \Omega \rightarrow \mathbb{R}$ be defined by

$$f(\mathbf{x}) = \log \det (A_0 + x_1 A_1 + \dots + x_n A_n)$$

Note that f is well-defined on Ω since the determinant of a positive definite matrix is positive.

Then $f = h \circ g$ is the composition of the affine function $g : \Omega \rightarrow \mathbb{S}^p$ defined by

$$g(\mathbf{x}) = A_0 + x_1 A_1 + \cdots + x_n A_n$$

with $h : \mathbb{S}_{++}^p \rightarrow \mathbb{R}$ by $X \mapsto \log \det X$.

We just found that

$$[Dh(X)](H) = \text{tr}(X^{-1}H)$$

and clearly (exercise if it's not clear)

$$[Dg(\mathbf{x})](\mathbf{h}) = h_1 A_1 + \cdots + h_n A_n$$

By the chain rule, $D(h \circ g)(\mathbf{x}) = Dh(g(\mathbf{x})) \circ D(g(\mathbf{x}))$ and

$$\begin{aligned}
 [Dh(g(\mathbf{x})) \circ D(g(\mathbf{x}))](\mathbf{h}) &= [Dh(g(\mathbf{x}))]([D(g(\mathbf{x}))](\mathbf{h})) \\
 &= \text{tr} (g(\mathbf{x})^{-1} [D(g(\mathbf{x}))](\mathbf{h})) \\
 &= \text{tr} (g(\mathbf{x})^{-1} (h_1 A_1 + \cdots + h_n A_n)) \\
 &= h_1 \text{tr} (g(\mathbf{x})^{-1} A_1) + \cdots + h_n \text{tr} (g(\mathbf{x})^{-1} A_n) \\
 &= \begin{bmatrix} \text{tr} (g(\mathbf{x})^{-1} A_1) \\ \vdots \\ \text{tr} (g(\mathbf{x})^{-1} A_n) \end{bmatrix}^\top \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix} = \nabla f(\mathbf{x})^\top \mathbf{h}
 \end{aligned}$$

Hence

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \operatorname{tr} \left((A_0 + x_1 A_1 + \cdots + x_n A_n)^{-1} A_1 \right) \\ \vdots \\ \operatorname{tr} \left((A_0 + x_1 A_1 + \cdots + x_n A_n)^{-1} A_n \right) \end{bmatrix}$$

The Hessian: definition and chain rules for vector-variate functions

The Hessian for vector-variate functions

For a real-valued function $f : \Omega \rightarrow \mathbb{R}$ where $\Omega \subseteq \mathbb{U}$, the gradient of f is a $\mathbb{U}$ -valued function $\nabla f : \Omega \rightarrow \mathbb{U}$

This means that we can talk about its derivative $D(\nabla f)$ at any point $u \in \Omega$, which is by definition a linear map

$$D(\nabla f)(u) : \mathbb{U} \rightarrow \mathbb{U}$$

This map takes any $h \in \mathbb{U}$ to some $[D(\nabla f)(u)](h) \in \mathbb{U}$.

The **Hessian** is just another name for this derivative, i.e., the Hessian is the derivative of the gradient, and it is denoted

$$\nabla^2 f(u) := D(\nabla f)(u)$$

We stress that

- i. a Hessian is only defined for a real-valued function, and
- ii. it is a linear map.

As usual the special cases $U = \mathbb{R}^n$ and $\mathbb{R}^{m \times n}$ (or $\mathbb{S}^n$) are what matters most to us.

Definition of the Hessian for vector-variate functions

If our function is real-valued and takes vector inputs, then the Hessian is quite simple: it is just the matrix of second partial derivatives.

Formally:

Let $\Omega \subseteq \mathbb{R}^n$ be open and $f : \Omega \rightarrow \mathbb{R}$ be twice continuously differentiable at $\mathbf{x} \in \Omega$.

The gradient is a function $\nabla f : \Omega \rightarrow \mathbb{R}^n$.

The Hessian of f at $\mathbf{x}$ is then defined to be the derivative of the function ∇f at $\mathbf{x}$, i.e.,

$$\nabla^2 f(\mathbf{x}) := D(\nabla f)(\mathbf{x})$$

By definition $\nabla^2 f(\mathbf{x})$ is $D(\nabla f)(\mathbf{x})$, i.e., a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^n$

As we saw from the previous lecture, the only matrix we can put in the box below

$$[D(\nabla f)(\mathbf{x})](\mathbf{h}) = \square \mathbf{h}$$

has to be the Jacobian of ∇f , which is given by the dumb formula

$$J(\nabla f)(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) \right]_{i,j=1}^n \in \mathbb{R}^{n \times n}$$

But this only works when the derivative of the gradient is a matrix – which is true for real-valued functions of vector variables.

This is why people often refer to this matrix $J((\nabla f)(\mathbf{x}))$ as the Hessian of f . If all we are discussing are real-valued functions of vector variables, it would have been enough to simply take

$$\nabla^2 f(\mathbf{x}) = J(\nabla f)(\mathbf{x}) \quad (\text{a matrix})$$

instead of the more general definition

$$\nabla^2 f(\mathbf{x}) = D(\nabla f)(\mathbf{x}) \quad (\text{a linear map})$$

Why we won't be able to use the matrix of second partial derivatives for matrix-variate functions

The problem is that when we have real-valued functions of matrix variables, the gradient is $\nabla f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$

You can't represent a linear map $\mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ as a matrix without making some unnatural identifications that have implications when it comes to computations. This means that " $J(\nabla f)(X)$ " makes no sense.

Chain rules for Hessians for vector-variate functions

Chain rule 1: Composition with scalar-valued functions

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$.

Then

$$\underbrace{\nabla^2(g \circ f)(\mathbf{x})}_{\text{matrix}} = \underbrace{g'(f(\mathbf{x}))}_{\text{scalar}} \underbrace{\nabla^2 f(\mathbf{x})}_{\text{matrix}} + \underbrace{g''(f(\mathbf{x}))}_{\text{scalar}} \underbrace{\nabla f(\mathbf{x}) \nabla f(\mathbf{x})^\top}_{\text{matrix}}$$

This rule shows that when composing with a scalar function, the Hessian includes both the original curvature (scaled by g') and an additional term dependent on the second derivative of g .

This second term represents the interaction between changes in different directions.

Chain rule 2: Composition with affine maps

Let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ and $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Define $g : \mathbb{R}^n \rightarrow \mathbb{R}$ by $g(\mathbf{x}) = f(A\mathbf{x} + \mathbf{b})$. Then

$$\underbrace{\nabla^2 g(\mathbf{x})}_{n \times n} = \underbrace{A^\top}_{n \times m} \underbrace{\nabla^2 f(A\mathbf{x} + \mathbf{b})}_{m \times m} \underbrace{A}_{m \times n}$$

This rule shows how the Hessian transforms under linear changes of variables.

These can be proved using the earlier results for derivative and gradient.

For example the first case follows from applying the gradient chain rule for “composition with scalar-valued function” and then apply the product rule.

The second term in the result on the right-hand side is called a “rank-one matrix” and comes from the outer product of two vectors.

Hessian for matrix-variate functions

A pause for some definitions: Inner and outer product

Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. The *inner product* gives a scalar measure of similarity between $\mathbf{a}$ and $\mathbf{b}$. It is defined as

$$\mathbf{a}^\top \mathbf{b} = a_1 b_1 + \cdots + a_n b_n \in \mathbb{R}.$$

The *outer product* is a matrix that captures all possible products between components of the two vectors. It is defined as

$$\mathbf{a}\mathbf{b}^\top = \begin{bmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & a_2 b_2 & \cdots & a_2 b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n b_1 & a_n b_2 & \cdots & a_n b_n \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

A special case of the outer product is $\mathbf{a}\mathbf{a}^\top = [a_i a_j]_{i,j=1}^n \in \mathbb{S}^n$.

Viewed another way, for $f : \mathbb{R}^n \rightarrow \mathbb{R}$, if we regard $\nabla^2 f(\mathbf{x})$ as $J(\nabla f)(\mathbf{x})$, then the latter is the matrix we can put in the box below so that

$$\lim_{\mathbf{h} \rightarrow 0} \frac{\|\nabla f(\mathbf{x} + \mathbf{h}) - \nabla f(\mathbf{x}) - \square \mathbf{h}\|}{\|\mathbf{h}\|} = 0$$

holds.

We may in fact define ∇f and $\nabla^2 f$ simultaneously! They are the vector and matrix we can put respectively in the first and second boxes below so that

$$\lim_{\mathbf{h} \rightarrow 0} \frac{|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \square^\top \mathbf{h} - \frac{1}{2} \mathbf{h}^\top \square \mathbf{h}|}{\|\mathbf{h}\|^2} = 0$$

$$\lim_{\mathbf{h} \rightarrow 0} \frac{|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \nabla^2 f(\mathbf{x}) \mathbf{h}|}{\|\mathbf{h}\|^2} = 0$$

This connects the Hessian to the Taylor series expansion of a function.

For the univariate case, the second order Taylor series expansion is

$$f(x + h) \approx f(x) + f'(x)h + \frac{1}{2}f''(x)h^2$$

For the multivariate case, the second order Taylor series expansion is

$$f(\mathbf{x} + \mathbf{h}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{h} + \frac{1}{2} \mathbf{h}^T (\nabla^2 f(\mathbf{x})) \mathbf{h}$$

There is no dumb formula for the Hessian of a matrix-variate function, because there is no Jacobian for matrix-variate functions.

Let $\Omega \subseteq \mathbb{R}^{m \times n}$ be open and $f : \Omega \rightarrow \mathbb{R}$ be twice continuously differentiable at $X \in \Omega$.

Note that the gradient is a function $\nabla f : \Omega \rightarrow \mathbb{R}^{m \times n}$. The Hessian of f at X is then defined to be the derivative of the function ∇f at X , i.e.,

$$\nabla^2 f(X) := D(\nabla f)(X)$$

Note that again $\nabla^2 f(X)$ is a linear map $\nabla^2 f(X) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$.

You will see later why it's important to define Hessian in this manner. The same definition applies if we have $\mathbb{S}^n$ in place of $\mathbb{R}^{m \times n}$.

Examples of Hessians of vector-variate functions

Example 1: $f = \frac{1}{x_1 x_2}$

Find the Hessian of $f(x_1, x_2) = \frac{1}{x_1 x_2}$:

This is a simple formula, and it is a real-valued function of vector variables, so we can use the dumb formula: the Hessian is the Jacobian of the gradient.

We can calculate the gradient too using the dumb formula:

$$\nabla f(x_1, x_2) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(x_1, x_2) \\ \frac{\partial f}{\partial x_2}(x_1, x_2) \end{bmatrix} = \begin{bmatrix} -\frac{1}{x_1^2 x_2} \\ -\frac{1}{x_1 x_2^2} \end{bmatrix}$$

The Hessian is the Jacobian of the gradient:

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(x_1, x_2) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(x_1, x_2) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(x_1, x_2) & \frac{\partial^2 f}{\partial x_2^2}(x_1, x_2) \end{bmatrix}$$

$$\nabla^2 f(x_1, x_2) = \frac{1}{x_1 x_2} \begin{bmatrix} 2/x_1^2 & 1/(x_1 x_2) \\ 1/(x_1 x_2) & 2/x_2^2 \end{bmatrix}$$

Example 2: $f = \frac{x_1}{x_2}$

Hessian of $f(x_1, x_2) = \frac{x_1}{x_2}$ is

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} 0 & -1/x_2^2 \\ -1/x_2^2 & 2x_1/x_2^3 \end{bmatrix}$$

Example 3: $f = \frac{1}{2}\mathbf{x}^\top A\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$

The definition is often easier to use than dumb formula.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be given by

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top A\mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$$

where $A \in \mathbb{S}^n$ and $\mathbf{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

Then since

$$f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) = (A\mathbf{x} - \mathbf{b})^\top \mathbf{h} + \frac{1}{2}\mathbf{h}^\top A\mathbf{h}$$

we see that

$$\nabla f(\mathbf{x}) = A\mathbf{x} + \mathbf{b}, \quad \nabla^2 f(\mathbf{x}) = A$$

Recall that if A is not symmetric then as

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top \left(\frac{A + A^\top}{2} \right) \mathbf{x}$$

we get

$$\nabla f(\mathbf{x}) = \frac{1}{2} (A + A^\top) \mathbf{x} + \mathbf{b}, \quad \nabla^2 f(\mathbf{x}) = \frac{1}{2} (A + A^\top)$$

Example 4: Linear Fractional Transformation

Sometimes chain rule can be very helpful. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined by

$$f(\mathbf{x}) = \frac{\|A\mathbf{x} + \mathbf{b}\|^2}{\mathbf{c}^\top \mathbf{x} + d}$$

Define $g : \mathbb{R}^n \times (0, \infty) \rightarrow \mathbb{R}$ by

$$g(\mathbf{y}, t) = \frac{\mathbf{y}^\top \mathbf{y}}{t} = \frac{\|\mathbf{y}\|^2}{t}$$

Then

$$\nabla^2 g(\mathbf{y}, t) = 2 \begin{bmatrix} \frac{1}{t} I_n & -\frac{\mathbf{y}}{t^2} \\ -\frac{\mathbf{y}^\top}{t^2} & \frac{\mathbf{y}^\top \mathbf{y}}{t^3} \end{bmatrix} \in \mathbb{S}^{n+1}$$

This representation is given as an block matrix or partitioned matrix:

$$\left[\begin{array}{c|c} \mathbf{M} & \mathbf{c} \\ \hline \mathbf{r} & s \end{array} \right],$$

$\mathbf{M}$ matrix $\mathbf{c}$ column vector $\mathbf{r}$ row vector s scalar Note that matrix-vector multiply works as if the block 2×2 matrix is really a 2×2 matrix

$$\begin{bmatrix} \mathbf{M} & \mathbf{c} \\ \mathbf{r}^\top & s \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ t \end{bmatrix} = \begin{bmatrix} \mathbf{M}\mathbf{x} + t\mathbf{c} \\ \mathbf{r}^\top\mathbf{x} + s \end{bmatrix}$$

Notation 9.4 (Diag operator). Let $\mathbf{a} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$. We write

$$\text{diag}(\mathbf{a}) = \begin{bmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{bmatrix} \in \mathbb{R}^{n \times n}, \quad \text{diag}(A) = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n$$

Exercise 9.5. Find $\nabla^2 f(\mathbf{x})$. As a hint, let $\mathbf{y} = A\mathbf{x} + \mathbf{b}$ and $t = \mathbf{c}^\top \mathbf{x} + d$ in $g(\mathbf{y}, t)$.

Example 5: Log-sum-exp or LSE function

Define $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$f(\mathbf{x}) = \log \left(\sum_{i=1}^m e^{\mathbf{a}_i^\top \mathbf{x} + b_i} \right)$$

where $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{R}^n$ and $b_1, \dots, b_m \in \mathbb{R}$. Write

$$A = \begin{bmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_m^\top \end{bmatrix} \in \mathbb{R}^{m \times n}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

so that

$$\begin{bmatrix} \mathbf{a}_1^\top \mathbf{x} + b_1 \\ \vdots \\ \mathbf{a}_m^\top \mathbf{x} + b_m \end{bmatrix} = A\mathbf{x} + \mathbf{b}$$

Define $g : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$g(\mathbf{y}) = \text{lse}(\mathbf{y}) := \log \left(\sum_{i=1}^m e^{y_i} \right)$$

Then $f(\mathbf{x}) = g(A\mathbf{x} + \mathbf{b})$. The gradient and Hessian of g are given by

$$\nabla g(\mathbf{y}) = \frac{1}{\sum_{i=1}^m e^{y_i}} \nabla \left(\sum_{i=1}^m e^{y_i} \right) = \frac{1}{\sum_{i=1}^m e^{y_i}} \begin{bmatrix} e^{y_1} \\ \vdots \\ e^{y_m} \end{bmatrix}$$

$$\nabla^2 g(\mathbf{y}) = D(\nabla g)(\mathbf{y}) = \begin{bmatrix} \frac{\partial g}{\partial y_1} & & \\ & \ddots & \\ & & \frac{\partial g}{\partial y_n} \end{bmatrix} - \left[\frac{\partial g}{\partial y_i} \frac{\partial g}{\partial y_j} \right]_{i,j=1}^m$$

Notation 9.7. If $\mathbf{a} \in \mathbb{R}^n$, then $\text{diag}(\mathbf{a}) \in \mathbb{S}^n$ is the diagonal matrix with i th entry a_i . More compactly,

$$\nabla^2 g(\mathbf{y}) = \underbrace{\text{diag}(\nabla g(\mathbf{y}))}_{n \times n} - \underbrace{[\nabla g(\mathbf{y})]}_{n \times 1} \underbrace{[\nabla g(\mathbf{y})]^\top}_{1 \times n}$$

By the chain rule (note that the roles of f and g have been reversed in the way we stated chain rule for Hessian),

$$\begin{aligned} \nabla^2 f(\mathbf{x}) &= A^\top \nabla^2 g(A\mathbf{x} + \mathbf{b}) A = \\ &A^\top \left[\left(\frac{1}{\sum_{i=1}^m e^{\mathbf{a}_i^\top \mathbf{x} + b_i}} \right) \begin{bmatrix} e^{\mathbf{a}_1^\top \mathbf{x} + b_1} & & \\ & \ddots & \\ & & e^{\mathbf{a}_m^\top \mathbf{x} + b_m} \end{bmatrix} \right. \\ &\quad \left. - \frac{1}{(\sum_{i=1}^m e^{\mathbf{a}_i^\top \mathbf{x} + b_i})^2} \begin{bmatrix} e^{\mathbf{a}_1^\top \mathbf{x} + b_1} \\ \vdots \\ e^{\mathbf{a}_m^\top \mathbf{x} + b_m} \end{bmatrix} \begin{bmatrix} e^{\mathbf{a}_1^\top \mathbf{x} + b_1} \\ \vdots \\ e^{\mathbf{a}_m^\top \mathbf{x} + b_m} \end{bmatrix}^\top \right] \end{aligned}$$

Examples: Hessians of matrix-variate functions

Example 1: log determinant

The next example is enormously important in semidefinite programming (SDP).

Let $f : \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ defined by $f(X) = \log \det X$. As we have shown before, its gradient is the matrix-valued, matrix-variate function

$$\nabla f : \mathbb{S}_{++}^n \rightarrow \mathbb{S}^n, \quad \nabla f(X) = X^{-1}$$

To find the Hessian, we need to differentiate the matrix-valued function of a matrix variable $X \mapsto X^{-1}$. Recall that for $\|A\| < 1$,

$$(I + A)^{-1} = (I - (-A))^{-1} = \sum_{i=0}^{\infty} (-A)^i.$$

Since $H \rightarrow 0$, we will eventually have $\|X^{-1}H\| < 1$, so we may apply this below.

Let $H \in \mathbb{S}^n$. Then

$$\begin{aligned}\nabla f(X + H) &= (X + H)^{-1} = [X(I + X^{-1}H)]^{-1} \\ &= \left[I - X^{-1}H + (X^{-1}H)^2 - (X^{-1}H)^3 + (X^{-1}H)^4 - \dots \right] X^{-1} \\ &= X^{-1} - X^{-1}HX^{-1} + X^{-1}H \left[I - X^{-1}H + (X^{-1}H)^2 - \dots \right] X^{-1}HX^{-1} \\ &= X^{-1} - X^{-1}HX^{-1} + X^{-1}H(I + X^{-1}H)^{-1}X^{-1}HX^{-1}.\end{aligned}$$

We claim that $[\nabla^2 f(X)](H) = [D(\nabla f)(X)](H) = -X^{-1}HX^{-1}$. This follows from

$$\begin{aligned} \frac{\|\nabla f(X+H) - \nabla f(X) + X^{-1}HX^{-1}\|}{\|H\|} &= \frac{\|X^{-1}H(I + X^{-1}H)^{-1}X^{-1}HX^{-1}\|}{\|H\|} \\ &\leq \frac{\|X^{-1}\|^3 \|H\|^2 \|(I + X^{-1}H)^{-1}\|}{\|H\|} \\ &= \|X^{-1}\|^3 \|H\| \|(I + X^{-1}H)^{-1}\| \rightarrow 0 \end{aligned}$$

as $H \rightarrow 0$. So the Hessian of f is the linear map

$$\nabla^2 f(X) : \mathbb{S}^n \rightarrow \mathbb{S}^n, \quad H \mapsto -X^{-1}HX^{-1}$$

There is an alternative proof (that I find harder to remember). Start from

$$(X + H) - X = H,$$

multiply on the left by X^{-1} and on the right by $(X + H)^{-1}$ and rearrange to get

$$\begin{aligned}(X + H)^{-1} - X^{-1} &= -X^{-1}H(X + H)^{-1} \\ &= -X^{-1}HX^{-1} + X^{-1}H[X^{-1} - (X + H)^{-1}]\end{aligned}$$

where the second expression follows by adding and subtracting $X^{-1}HX^{-1}$.

Therefore,

$$\begin{aligned}\frac{\|(X+H)^{-1} - X^{-1} + X^{-1}HX^{-1}\|}{\|H\|} &= \frac{\|X^{-1}H[X^{-1} - (X+H)^{-1}]\|}{\|H\|} \\ &\leq \frac{\|X^{-1}\| \|H\| \|X^{-1} - (X+H)^{-1}\|}{\|H\|} \\ &= \|X^{-1}\| \|X^{-1} - (X+H)^{-1}\| \rightarrow 0\end{aligned}$$

as $H \rightarrow 0$.

Example 2: Trace inverse

We will do another example to illustrate the use of the chain rule.

Let $f : \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ defined by $f(X) = \text{tr}(X^{-1})$. As we saw before, its gradient is the matrix-valued, matrix-variate function

$$\nabla f : \mathbb{S}_{++}^n \rightarrow \mathbb{S}^n, \quad \nabla f(X) = -X^{-2}$$

To find the Hessian, we need to differentiate the matrix-valued function of a matrix variable $X \mapsto -X^{-2}$.

Let $g : \mathbb{S}_{++}^n \rightarrow \mathbb{S}^n$ be given by $g(X) = -X^{-1}$ and $h : \mathbb{S}_{++}^n \rightarrow \mathbb{S}_{++}^n$ be given by $h(X) = X^2$. Then clearly $\nabla f = g \circ h$. We know from the above example that

$$[Dg(X)](H) = X^{-1}HX^{-1}$$

and from an earlier lecture that

$$[Dh(X)](H) = XH + HX.$$

By the chain rule,

$$\begin{aligned}[D(g \circ h)(X)](H) &= [Dg(h(X))Dh(X)](H) \\ &= h(X)^{-1}([Dh(X)](H))h(X)^{-1} \\ &= X^{-2}([Dh(X)](H))X^{-2} \\ &= X^{-2}(XH + HX)X^{-2} \\ &= X^{-1}HX^{-2} + X^{-2}HX^{-1}.\end{aligned}$$

So the Hessian of f is the linear map

$$\nabla^2 f(X) : \mathbb{S}^n \rightarrow \mathbb{S}^n, \quad H \mapsto X^{-1}HX^{-2} + X^{-2}HX^{-1}$$

Last words about multivariate derivatives

When do notions like gradient and Jacobian apply?

One point that we have been making implicitly is that notions like gradient and Jacobian (and Hessian defined as Jacobian of gradient) only apply in cases when the derivative, which is a linear map, can be represented as a matrix. The number of such cases is very limited:

Case 1: $f : \mathbb{R}^n \rightarrow \mathbb{R}$

$f : \mathbb{R}^n \rightarrow \mathbb{R}$: gradient $\nabla f(\mathbf{x}) \in \mathbb{R}^n$ is such that

$$[Df(\mathbf{x})](\mathbf{h}) = \nabla f(\mathbf{x})^\top \mathbf{h}$$

Case 2: $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$

$f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$: gradient $\nabla f(X) \in \mathbb{R}^{m \times n}$ is such that

$$[Df(X)](H) = \text{tr}(\nabla f(X)^\top H)$$

Case 3: $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$: Jacobian $Jf(\mathbf{x}) \in \mathbb{R}^{m \times n}$ is such that

$$[Df(\mathbf{x})](\mathbf{h}) = Jf(\mathbf{x})\mathbf{h}$$

$f : \mathbb{R}^n \rightarrow \mathbb{R}$: Hessian $\nabla^2 f(\mathbf{x}) := J(\nabla f)(\mathbf{x}) \in \mathbb{R}^{n \times n}$ is such that

$$[D(\nabla f)(\mathbf{x})](\mathbf{h}) = \nabla^2 f(\mathbf{x})\mathbf{h}$$

This fails in more general scenarios. We cannot define a ‘Jacobian’ for $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$ as a matrix which why there is no such notion.

We also cannot define a Hessian for $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ in terms of a matrix which is why we need to define it directly in terms of derivative.

The notion of derivative is the one that works under all circumstances - because a derivative is a linear map, not a matrix.

In linear algebra, you learned that every linear map can be represented by matrices but this is a useless statement.

What you want is for the linear map to be not only represented by matrices but to have an expression as one single matrix-vector or matrix-matrix product. There are only three cases where you can do this:

Case 1: linear maps from $\mathbb{R}^n \rightarrow \mathbb{R}^m$

linear maps from $\mathbb{R}^n \rightarrow \mathbb{R}^m$ can be represented as

$$\mathbb{R}^n \rightarrow \mathbb{R}^m, \quad \mathbf{x} \mapsto A\mathbf{x}$$

for some matrix $A \in \mathbb{R}^{m \times n}$;

Case 2: linear maps from $\mathbb{R}^n \rightarrow \mathbb{R}$

linear maps from $\mathbb{R}^n \rightarrow \mathbb{R}$ can be represented as

$$\mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathbf{x} \mapsto \mathbf{a}^\top \mathbf{x}$$

for some vector $\mathbf{a} \in \mathbb{R}^n$;

Case 3: linear maps from $\mathbb{R}^{m \times n} \rightarrow \mathbb{R}$

linear maps from $\mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ can be represented as

$$\mathbb{R}^{m \times n} \rightarrow \mathbb{R}, \quad X \mapsto \operatorname{tr}(A^T X)$$

for some matrix $A \in \mathbb{R}^{m \times n}$. If you have, say, a linear map $\mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ like $X \mapsto AX + XB$, you can no longer write it down as one single matrix-matrix product without introducing some artificial notions.